Yang Xu

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Education

Zhejiang University Hangzhou, China | 2022–2027 (expected)
Undergraduate, Computer Science and Technology; Advanced Honor Class for Engineering Education, Chu Kochen Honors College (50 science and engineering students selected per cohort). GPA: 3.99/4.3.

Research Interests

I believe that scaling is a science. I want to build its experimental instruments as carefully as a physicist builds a cyclotron, so that each training run tests a real hypothesis and teaches us something that survives at the next scale. I study new model architectures, from attention to depth, together with their scaling laws and the transfer of hyperparameters across scales. My earlier work in interpretability and trustworthy machine learning taught me to ask what changes inside a model when an experiment works, and why. My near-term goal is to use models to accelerate the science of scaling itself, and to use what we learn from scaling to train models capable of scientific reasoning and discovery.

Selected Work

* means equal contribution.

MiniMax Sparse Attention. arXiv:2606.13392.
Xunhao Lai, Weiqi Xu, Yufeng Yang, Qiaorui Chen, **Yang Xu**, Lunbin Zeng, Xiaolong Li, Haohai Sun, Haichao Zhu, Vito Zhang, Jinkai Hu, Jiayao Li, Rui Gao, Zekun Li, Songquan Zhu, Jingkai Zhou, Pengyu Zhao.

MiniMax M3: Frontier Coding, 1M Context, Native Multimodality — All in One Model.
MiniMax Team. Contributed to pretraining, model architecture, and algorithm–engineering co-design.

LH-Deception: Simulating and Understanding LLM Deceptive Behaviors in Long-Horizon Interactions. ICLR 2026.
Yang Xu*, Xuanming Zhang*, Samuel Yeh, Jwala Dhamala, Ousmane Dia, Rahul Gupta, Sharon Li.

Tracking the Feature Dynamics in LLM Training: A Mechanistic Study. arXiv:2412.17626.
Yang Xu, Yi Wang, Hengguan Huang, Hao Wang.

Towards the Dynamics of a DNN Learning Symbolic Interactions. NeurIPS 2024.
Qihan Ren*, Junpeng Zhang*, **Yang Xu**, Yue Xin, Dongrui Liu, Quanshi Zhang.

Layerwise Change of Knowledge in Neural Networks. ICML 2024.
Xu Cheng*, Lei Cheng*, Zhaoran Peng, **Yang Xu**, Tian Han, Quanshi Zhang.

Experience

Principia Labs | San Francisco, USA (remote from Hangzhou, China) 2026.07–Present
Foundation-model pretraining, architecture experiments, and scalable, reproducible training systems.

NonLinear-1 | Online 2026.06–Present
Organizer of a research group for open scientific discussion and LLM training.
Compute resources backed by Starting Point One.

MiniMax | Shanghai, China 2026.02–2026.07
Algorithm and engineering co-design for large-language-model pretraining and scaling.

Summer Research at N1 | Beijing, China 2025.07–2025.08
Mentored by Jingyang Yuan; Triton kernel optimization and efficient attention.

UW–Madison | Madison, USA (remote) 2025.03–2025.11
Advised by Prof. Sharon Li; LLM trustworthiness evaluation in collaboration with Amazon.

Rutgers University | New Jersey, USA 2024.07–2025.02
Advised by Hao Wang (now Associate Professor at UIUC); mechanistic interpretability of LLMs.

Shanghai Jiao Tong University | Shanghai, China (remote) 2023.05–2024.09
Advised by Prof. Quanshi Zhang. Neural-network interpretability and theory.

Academic Service

Reviewer: ICLR 2025, NAACL 2025, and ICLR 2026.